When Sigmoid Calibration Worsens Multiclass Logistic Regression: Effects of Calibration Set Size and Score Representation

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Abstract

The post hoc calibration process can potentially enhance the accuracy of the predicted probabilities. However, this could lead to diminished baseline performance because of the addition of yet another fitted transformation. The current research provides both exploratory benchmarks and controlled follow-ups of sigmoid-calibrated multi-class logistic regression. Logistic regression, random forest and Gaussian Naive Bayes are tested on two data sets (breast cancer and handwritten digits) through 10 stratified partitions. There is a marked increase in the 10-bin expected confidence calibration error for logistic regression of the digits data set, from .0240 to .1332. In the follow-up portion, the fitted models, along with their corresponding test partitions, remain constant while the sample size for the calibration is varied through 20 additional partitions on three different data sets (digits, iris, and wine). Four methods are compared, utilizing the original probability estimates, applying sigmoid calibration of class logits, applying temperature scaling and finally applying sigmoid calibration of class-versus-rest log odds. In each subsequent comparison, the standard sigmoid log loss degrades. The average degradation for digits drops from .4906 (36 calibration observations) down to .1277 (360 calibration observations). As a consequence of using large amounts of calibration data, for the latter case, converting logits to log odds yields a deterioration of .0350 relative to baseline and temperature scaling yields a mean improvement of .0013. Furthermore, temperature scaling produces unstable log loss with scarce calibration data and conflicting metric changes on the smaller datasets. Overall, these findings show the value of having sufficient numbers of calibration observations in addition to the proper characterization of the scores, however they do not identify all possible distortions nor produce true population calibration. The contributions provided here are an empirical analysis which can be replicated and demonstrate insight into why a particular failure was encountered and not a novel calibration method or general guidance.

1 Introduction

Predictive classifiers have labels and/or numerically-provided probability vectors as output. A label is a declaration of a preferred outcome and a probability vector is a description of how preferred that preferred outcome is over its alternatives. Both types of output need to be evaluated differently. Models can correctly classify the correct class for most observations and create confidence levels that are inconsistent with empirical correctness. Conversely, models can supply more believable uncertainty bounds for misclassifying at a greater frequency. Therefore, applications that use numerical predictions to either prompt review or prioritize/abstain from classification should be accompanied with measures related to the quality of those probability predictions.

Probability Calibration is focused on evaluating whether predictions agree with actual outcomes. For example, if there exists sufficient amounts of identical predictions with a 90% confidence level then approximately 90% of those predictions will be accurate. This is based on a long-run relationship and does not ensure that any single prediction will be accurate. The type(s) of probabilities being calibrated (maximum confidence, probability of one specific class, etc.) cannot be quantitatively determined via any singular evaluation metric.

Evaluation needs to clearly state what target(s) are being evaluated upon and recognize that the finite samples represent only estimates of the probability distributions generated by the underlying data generating processes.

Post-hoc calibration refers to calibration performed after the predictive model has been trained. One advantage of post-hoc calibration is its utility: assuming there exists a collection of observations that were not included in the original training set for a classifier, an estimator for a probability transformation may be estimated without requiring retraining of the original classifier. However, assumptions and uncertainties associated with estimation are present during the transformation process. Assuming that an operation's goal is to perform some form of calibration then it would constitute a calibration operation; however, "guarantee" that such an operation would have a positive calibration effect is not supported. An original predictor that produces useful probabilities may produce lower-quality probabilities as a result of additional fitting or switch classes as a result of additional fitting or demand an inappropriate functional relationship among the predictors. Such reasons justify comparing against an alternative that represents the original model.

The initial experimentation involved employing three types of classifiers for two datasets. Sigmoid function addition improved the performance of Gaussian Naive Bayes on both tasks and random forest on digits while reducing the performance of logistic regression.

The most surprising part of the logistic regression on the digits dataset was how far the increase in confidence error exceeded the reduction in accuracy. Based on this difference, a more specific question was asked about the effects of the size of the calibration set and choice of multi-class calibration method on the quality of probabilities produced by logistic regression.

A further diagnostic question was if the score inputted into a class-wise sigmoid would be relevant. The experiment consists of two phases: the base-line phase and the follow-up phase, focusing specifically on investigating the results related to the logistic regression. The description of methods can be found in section 2, and the reporting of results is found in section 3.

Interpretations, limitations, and conclusions are discussed in sections 4-6. The first phase influenced the second design; therefore, the first phase is not presented as a confirmation in terms of pre-registration. Different follow-up split seed measurements assess partition sensitivity, but re-use of the digits dataset does not constitute external validation on newly collected observations.

Appendix A-D provide supporting definitions, figures, full results, and reproducibility information.

1.1 Related work

Comparative research clearly separated the accuracy of labeling and the quality of probabilities for many years. Researchers (Niculescu-Mizil & Caruana, 2005) compared supervised learners and post-hoc corrections to probabilities. Guo et al. (2017), in their exploration of neural networks' calibration, found that temperature scaling performed well in the cases that they investigated.

The evidence is being utilized to motivate a simple multiclass comparator (and not that a scaling similar to temperature should inherently improve classical logistic regression or be reliable when working with very small calibration sets).

Additionally, the possibility for a negative correction is also shown. Kull et al. (2017) showed that logistic calibration is limited and developed beta calibration for binary predictions. The authors' identity function argument relates to specific input values and transformations; and, should not simply be passed down to sigmoid fitting on logits.

Kull et al. (2019) suggested a native multiclass alternative, Dirichlet calibration. Ojeda et al. (2023) examined probability recalibration through numerous simulations with varying sample sizes and other variables. However, neither recalibration failure nor sensitivity to sample size is mentioned anywhere within the current paper as a novel general discovery.

The smaller contribution is a more precise diagnosis to the current implementation that provides support via matched comparisons. It provides a link between an easy-to-understand measure of failure and deliberated changes to sample size, transformation families, and representations of the scores.

This is defined as a narrowly-stated purpose. The study does not claim that this phenomenon was absent from prior investigations; there is no creation of an entirely new estimator; and, there is no comprehensive algorithmic comparison.

Its merit lies in providing clarity as to a previously observed reconstruction effect and determining among various possible explanations which may have remained speculative.

2 Methods

2.1 Preliminary benchmark

In this project, two sets of benchmarking data are utilized. These include: a) The Wisconsin Diagnostic Breast Cancer Dataset; b) Handwritten digits dataset of scikit-learn.

This includes 569 breast cancer cases with 30 numerical features and two diagnostic labels. In addition, there are 64 feature values in 1797 observations in the Digit case, which are comprised of 8x8 images labeled with ten classes of digits. These are examples of handwriting and recorded measurements, not artificially created. As a result, the analysis will be conducted using a subset of the existing benchmark measurement and digit image data, without involving additional subjects or data collection methods (Alpaydin and Kaynak, 1998; Wolberg et al., 1993).

Stratified sampling is used to create stratified partitions of the data with seeds ranging from 0 to 9. Approximately thirty percent of the total data is held out for testing. The remaining development data includes approximately fifty percent for model building and twenty percent for calibration. The quantities for the number of observations for which you wish to get exact counts are: The breast-cancer data: 284 for model building, 114 for calibration and 171 for test observations; The digit data: 897, 360, and 540.

Stratification allows us to maintain approximately the original class proportions, but it doesn't allow us to take into consideration additional factors that could influence groupings, such as shared writers, collection sites, etc. that might exist in the data but weren't recorded.

We use standardized predictors, standard regularization and the max # of iterations = 3000 for solving logistic regression. For Random Forests, we use a fixed number of trees (100) and a random seed that is selected specifically for each partition. We also use default parameters for the Gaussian Naive Bayes models. Before we apply sigmoid calibration to the fitted models, we "freeze" the base classifiers. Thus, we don't retrain a new base classifier in order to calibrate them. We are able to assess performance before and after adjusting the predicted probabilities via repeated holdout evaluations.

The benchmark generates 120 rows: Two datasets, three classifiers, ten partitions and two variants. Therefore, the rows represent repeated assessments of the two datasets instead of representing 120 separate experiments.

2.2 Outcomes and mathematical notation

Let p_{ik} denote the estimated probability of class membership of the ith observation in class k, let y_i denote the true class of the ith observation and let $\hat{y}_i$ denote the predicted class with highest estimated probability for the ith observation. Finally, define $c_i = \max_k p_{ik}$ and $z_i = 1\{\hat{y}_i = y_i\}$. The indicator function is defined as follows: $z_i = 1$ if $\hat{y}_i = y_i$, and $z_i = 0$ otherwise. Here we concentrate on the maximal element of the label-probability vector, so while this term is often called confidence in various applications, it is important to note that it is not the estimated probability of malignancy when assessing the binary classification problem of determining whether a tumor is malignant or benign. Additionally, since this is a multi-class problem, we ignore the distribution of estimated probabilities among the non-winning classes.

$$\hat{y}_i = \arg\max_k p_{ik}, \quad c_i = \max_k p_{ik}, \quad \text{Accuracy} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{\hat{y}_i = y_i\} \tag{1}$$

Log Loss (LL): This evaluates how well your model estimates the probability of observing the correct class. Brier Score (BS): This adds up each class's squared error in the full probability vector and then averages over observations. Like LL, BS is a probability scoring rule and thus does not measure calibration directly. When evaluating differences between Brier Scores within a specific task (e.g., do not mix summed and single-event Brier scores) use consistent conventions; however, avoid considering values as generally applicable across multiple tasks with differing numbers of classes (Brier, 1950).

$$\text{LL} = -\frac{1}{n}\sum_{i=1}^{n} \log p_{i,y_i}, \qquad \text{BS} = \frac{1}{n}\sum_{i=1}^{n}\sum_{k=1}^{K} (p_{ik} - \mathbf{1}\{y_i = k\})^2 \tag{2}$$

Expected Calibration Error (ECE): ECE is a binned metric for measuring a model's marginal confidence calibration. ECE measures the average absolute difference between the confidence level provided by a model about its own predictive accuracy and its actual accuracy. Confidence levels fall into one of ten equal sized intervals over [0, 1]. Confidence levels associated with empty bins are simply ignored. Because ECE is a metric for overall confidence calibration, and it does not depend on predicted class, ECE cannot provide a class-wise assessment of calibration and it does not identify all class-wise failures. Vaicenavicius et al. (2019) describe why this distinction matters when analyzing such summaries.

$$\text{ECE}_B = \sum_{b=1}^{B} \frac{|I_b|}{n} \left| \frac{1}{|I_b|}\sum_{i \in I_b} c_i - \frac{1}{|I_b|}\sum_{i \in I_b} \mathbf{1}\{\hat{y}_i = y_i\} \right| \tag{3}$$

Means and Sample Standard Deviation: These statistics are used to compute variability between partitions. To evaluate partition-by-partition differences in score, we first calculate the score computed on the partition that was modified minus the original score. Negative changes indicate reduced probability-scoring error; positive changes in accuracy indicate increased accuracy. Due to our decision to restrict comparisons to directionally observable effects versus uncertainty concerning future deployments, repeated holdout averages include no statistical tests for significance and no population-based confidence intervals.

2.3 Controlled follow up design

The follow-up keeps digits and includes iris and wine. Iris has 150 observations, four features measured on those observations and three labels. There are 178 observations in wine, thirteen chemical features and three labels (Fisher, 1936; Aeberhard and Forina, 1992). These small datasets provide contrasting multiclass tasks with varying calibration budgets. They are not representative of application domains, but convenience benchmarks. They vary in the number of features, the number of samples, the balance of the classes and the difficulty of the classification making the between-dataset effects difficult to attribute to any one factor.

Partitions created by seeds 100-119 are 20 each per task. Both allocate around the same proportion of 50/20/30 as the benchmark. The logistic classifier (the standard one) is fitted once for each partition. The values of its fitting observations, learned parameters and test observations are the same for all follow-up methods and calibration sizes. This paired design is critical: Variation of the fitting-size and calibration-size would mask the improvement of the base classifier with improvement in the correction. In this case, the intervention alters the calibration of the models but does not alter the prediction problem that each of the models is tasked with solving.

Observations of the calibrated items are randomly ordered within each class and interleaved in class order. The prefixes of that sequence give sub-sets of the class with near equal number of elements. Target sizes are

10%, 25%, 50% and 100% of the reserved calibration pool with a minimum of 2 observations per target size class. Digits sizes are 36, 90, 180, and 360; iris sizes are 6, 8, 15, and 30; wine sizes are 6, 9, 18, and 36. The minimum is the reason that the smallest subset, both the iris and wine subsets are larger than their nominal 10%.

Round-robin sampling approximately balances early prefixes rather than reproducing exact population class proportions. The original stratified allocation will be restored when the pool is full size. Nested comparisons eliminate one source of comparison noise, but the design includes one ordering of calibrations per fitted classifier. It fails to estimate the variability of repeated subset selection with the same classifier. Several distinct calibration orderings would reinforce that decomposition. Twenty partitions measure a combination of the sensitivity to training, test allocation and calibration composition.

2.4 Calibration methods

Multinomial logistic regression assigns class scores s_{ik}, and uses the softmax function to calculate probabilities. The standard sigmoid calibration is a binary correction for each class score and then normalizes the positive values to a sum equal to one. Coefficients vary from class to class. The following is the scikit-learn (v1.8.0) version employed in the experiment. The fitted classifier is frozen, and a single prediction fold does not have to satisfy the minimum number of counts in each class, which is often a requirement of default 5-fold procedure. That fold does not refit the base model; calibration labels are used only to fit the correction.

$$p_{ik} = \frac{\exp(s_{ik})}{\sum_{j=1}^{K} \exp(s_{ij})} \tag{4}$$

The rescaling of temperature is performed before softmax, with only one positive temperature T. The parameter is chosen to minimize the calibration-set mean negative log likelihood. Optimisation is done for log T within [−4, 4], corresponding to approximately [0.0183, 54.60] for T. The bound is explicit so that there are no arbitrary large numerical excursions, but it does not ensure that the out-of-sample probabilities are stable. The allowed interval includes T = 1, which leaves the model unchanged. A positive common temperature keeps the class scores in the same order as in the original model, and as a result the winning label remains in place.

$$p_{ik}^{(T)} = \frac{\exp(s_{ik}/T)}{\sum_{j=1}^{K} \exp(s_{ij}/T)}, \qquad T > 0 \tag{5}$$

The "score-representations" ablation differs from the "standard" ablation, in that the log odds of the classes relative to their respective complements are utilized rather than the unnormalized class-scores. The original probabilities are clipped to $[10^{-12}, 1 - 10^{-12}]$ before converting to log odds. This ablation is NOT a recommended alternative calibration method. This is an exploratory diagnostic alternative.

$$r_{ik} = \log\left(\frac{p_{ik}}{1-p_{ik}}\right), \quad q_{ik} = \frac{1}{1+\exp[-(a_k r_{ik} + b_k)]}, \quad \tilde{p}_{ik} = \frac{q_{ik}}{\sum_j q_{ij}} \tag{6}$$

Further, the distinction between raw logits and log-odds becomes much more important when modeling multiple classes. The probability of each class depends on both its own class score, as well as how its class score relates to the class scores of every other class. As such, even though two observations might have the exact same value of s_{ik} , they may still have different probabilities p_{ik} based on which of the possible choices they have. These comparisons cannot be captured using a separate sigmoid of each raw class score before

normalization. In fact, these comparisons are preserved in the original probabilities, and the log-odds of each class against the rest of the classes reflect these comparisons. The algebra provided to support the ablation does not establish a clear connection between misrepresentation and increases in loss for all observed losses.

To simplify things further, let us define the normalizing transform for the standard model as $q_{ik} = \sigma(a_k\, s_{ik} + b_k)$. Then the normalized values are defined as $q_{ik} / \sum_j q_{ij}$. Classwise sigmoid models for K classes estimate 2 parameters per class (a and b), while temperature scaling models estimate one shared temperature. Parameter count may contribute to small-sample behavior, because temperature scaling estimates fewer parameters. However, the comparison of methods did not isolate parameter count, because both types of functions also differ. The targets we chose are $(n_+ +1)/(n_+ +2)$ for positive values and $1/(n_- +2)$ for negative values, where n_+ and n_- represent the positive and negative calibration counts for each class. Hence, our calibration budget is not only comprised of the estimation of the parameters, but also includes the fitting of the target itself. All remain within the log-odds ablation.

2.5 Follow up outcomes and verification

We use logarithmic loss (ll) or mean-negative-log-likelihood (NLL) as our main descriptive follow-up as it directly evaluates the probabilities assigned to the observed classes. In addition to ll, we assess summed Brier Score (bs), accuracy, mean-confidence and confidence ECE. We assess sensitivity to binning by generating ECE with 5, 10, and 20 equal-sized bins. Label Change Rate is a measure of the proportion of test observations for which the corrected vector has switched class from its original class. None of our tests utilize the outcome of any test to select a transformation to apply, modify any parameter(s) or select a particular partition to include.

All 960 evaluation rows for this study are created as follows: 3 Datasets * 20 Partitions * 4 Calibration Sizes * 4 Variants. In order to create matched tables, rather than having to generate 240 separate fits, we retain the original predictions in each of these rows. These repeated baseline rows represent sixty fitted classifiers, not 240 separate baseline fits. We preserve the information about splits, test indices, calibration orders, versions of environments and associated changes. Details concerning the data schema, computational environment, and files necessary to recreate this study are contained in Appendix D. We perform several checks to ensure that: the probabilities are finite; each row sums to 1; temperature labels do not change; we obtain the correct probabilities via computation of softmax; original probabilities and softmax match; and manually constructed sigmoid/logit predictions correspond to the public calibration API.

For purposes of making a consistent call across smoothing and optimization in the standard implementation, we used an internal version pinning sigmoid fitter for the log-odds diagnostic. If you wish to reproduce this study, this decision limits you to a specific private software interface for reproduction. It is documented in companion code, but we consider it non-generic and if the library changes in some way in the future, you should either reproduce or replace the fitting routine in that instance. The checks currently being performed were designed to verify that the implementations are numerically consistent, not to establish statistical validity for our assumptions.

3 Results

3.1 Preliminary benchmark and motivating observation

Table 1 Preliminary comparison of original and sigmoid probabilities

Dataset	Classifier	Original LL	Sigmoid LL	Original ECE	Sigmoid ECE	Δ accuracy
Breast cancer	Logistic regression	0.0795	0.0992	0.0232	0.0431	-0.0585
Breast cancer	Random forest	0.1798	0.1303	0.0373	0.0399	+0.0585
Breast cancer	Naive Bayes	0.6656	0.2197	0.0594	0.0268	+0.1754
Digits	Logistic regression	0.1264	0.2559	0.0240	0.1332	-0.5926
Digits	Random forest	0.3621	0.1381	0.2339	0.0621	+0.0185
Digits	Naive Bayes	2.9186	0.6857	0.1461	0.0725	+0.3519

LL = logarithmic loss; ECE = confidence expected calibration error with ten equal-width bins. Values are means across ten partitions. Δ accuracy is calibrated minus original in percentage points. Lower probability error scores are better. Complete accuracy, summed Brier scores, dispersion, and paired changes are in Tables B1–B5.

Table 1 summarizes the preliminary probability score comparison. Figure 1 includes reliability diagrams representing the logistic regression models that led to additional investigation. Figure 1 shows logistic regression only. Figures B1-B2 and Tables B6a-b show the original all-model plot, the separated panel representations, and the bin values.

An important goal of this research has been to assess the effectiveness of using a sigmoid transformation to adjust model predicted probabilities. The results reported in Table 1 show that the transformation was effective in lowering the average logarithmic loss (log-loss) of the Breast Cancer Naive Bayes Model from .6656 to .2197. It was also effective in lowering the expected calibration error (ECE) from .0594 to .0268. Likewise, the application of a sigmoid transformation to the Digits Naive Bayes Model lowered its log-loss from 2.9186 to .6857 and lowered ECE from .1461 to .0725. These results indicate a substantial improvement in probability scores, but little increase in the number of accurate labels generated by each model. This illustrates a distinction between improving probability quality and increasing the number of accurate labels generated by the model. However, these results do not provide insights regarding why the original models made erroneous classifications.

However, Digits Random Forest is an interesting example of how accuracy does not always correspond to confidence. While the estimated accuracy of the model is approximately 96.93%, there was substantial mismatch between confidence and observed accuracy. Specifically, the ECE was 0.2339. Following the application of the sigmoid transformation, accuracy remained relatively unchanged at 96.94%. However, the ECE decreased from 0.2339 to 0.0621 and log-loss decreased from 0.3621 to 0.1381. Thus, these results for Digits Random Forest suggest that high levels of label accuracy can exist along with large amounts of confidence mismatch. The improved confidence metrics do not establish how a downstream decision-maker would perceive these estimates or whether they would improve decisions.

For Breast Cancer Random Forest, the results were somewhat ambiguous. Log-loss was lowered from 0.1798 to 0.1303 after application of the sigmoid transformation. However, ECE was actually higher after application of the sigmoid transformation (.0399 compared to .0373). Additionally, only five out of ten splits resulted in improvements in log-loss. As mentioned previously, this discrepancy between a positive mean and lack of consistency at each split suggests that caution should be used when claiming a consistent improvement in probability quality. Additionally, the differences identified herein between probability scoring and confidence binning demonstrate that which aspects of a prediction will be emphasized will depend on whether you utilize one method versus another for assessing calibration. Even though disagreements may cause some

researchers to favor one metric over another for assessing calibration, it is most beneficial for understanding why they disagree.

Logistic regression probability scores worsened on both datasets. On breast cancer, mean log-loss changed from 0.0795 to 0.0992, and ECE changed from 0.0232 to 0.0431. On digits, log-loss increased from 0.1264 to 0.2559, Summed Brier Score increased from 0.0575 to 0.1089, and ECE increased from 0.0240 to 0.1332. Accuracy decreased from 96.43% to 95.83%. Since changes in magnitude of probability estimates offer a convenient means for diagnosing problems (see Figure 1), these results serve well as an illustration of how to perform such diagnostics

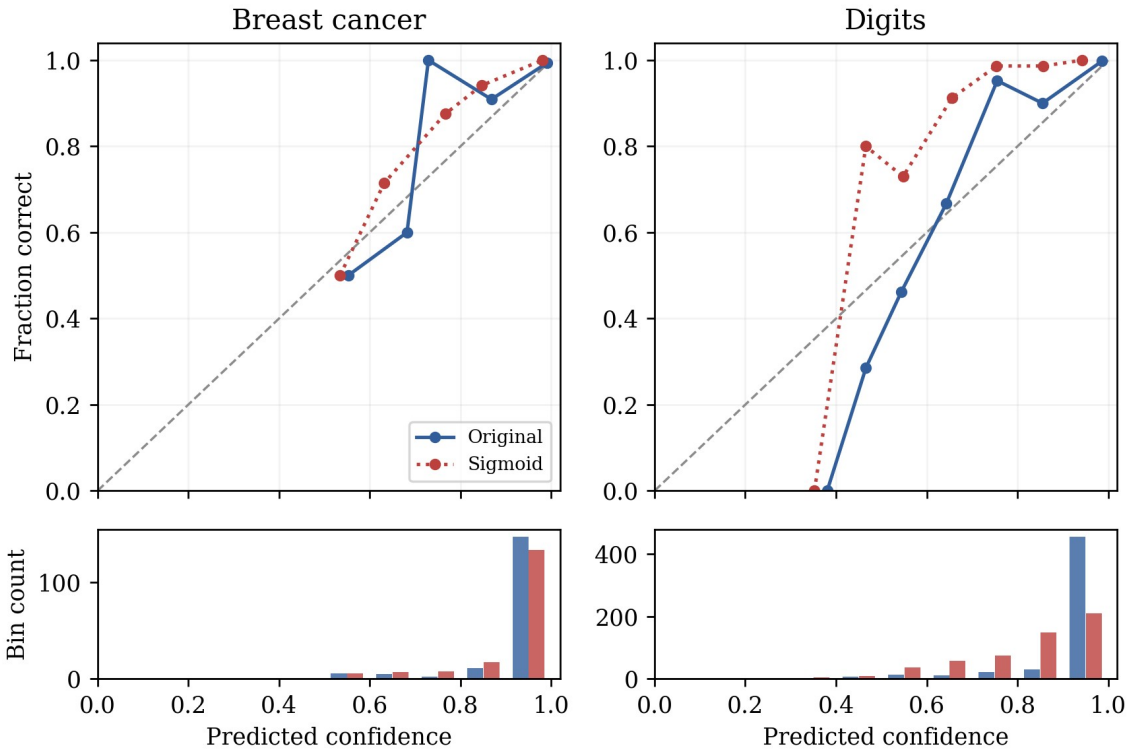


Figure 1 Original logistic regression reliability diagrams

Seed-zero confidence bins for breast cancer and digits. Original probabilities are blue solid lines; sigmoid-calibrated probabilities are red dotted lines. The grey diagonal represents agreement. Lower panels show bin counts at interval midpoints. These are single-partition illustrations, not averages across ten splits; all plotted bin coordinates and counts are in Tables B6a and B6b.

Although low observed-confidence ECE is enough to indicate that a small confidence gap in the observed digits logistic regression predictions; to make a claim that a particular population is "well-calibrated," you need to use significantly more sophisticated methods and independently verify the claims. The follow-up analysis does not select models based on a final-test threshold and makes no claim that all logistic classifiers are well-calibrated. Instead, the analysis examines an effective baseline probability measure and then investigates whether similar negative impacts occur when controlled changes are made to the baseline.

Iris and Wine provide a broader scope of experimental tasks beyond merely digits classification and provide an opportunity to utilize prior research supporting the feasibility of such experimentation.

3.2 Calibration size effects on digits

Table 2 Digits log loss across calibration sizes

n_{cal}	Original	Sigmoid logits	Sigmoid log odds	Temperature
36	0.1246 ± 0.0172	0.6152 ± 0.0335	0.4840 ± 0.0479	0.2983 ± 0.3121
90	0.1246 ± 0.0172	0.3858 ± 0.0171	0.2775 ± 0.0246	0.2141 ± 0.2513
180	0.1246 ± 0.0172	0.2991 ± 0.0136	0.1999 ± 0.0165	0.1260 ± 0.0211
360	0.1246 ± 0.0172	0.2523 ± 0.0165	0.1596 ± 0.0172	0.1233 ± 0.0193

n calibration = number of calibration observations. Values are mean ± sample standard deviation across twenty partitions. Each row retains the same fitted base classifiers and test observations; original scores therefore repeat across sizes.

Log-loss learning curves for digits are shown in Table 2. Learning curve data for Iris and Wine, which were examined separately from digits, are shown alongside digits' data in Figure 2. Additional data, including accuracy and confidence estimates at each calibration set size, are provided in Tables C1-C6.

The baseline digits classifier scores 96.36% accuracy, 0.1246 log loss, 0.0564 summed Brier score, and 0.0233 ten-bin ECE. These values are very close to the initial logistic regression benchmark. Using 36 calibration observations, standard sigmoid-logit calibration increases log loss to 0.6152 and ECE to 0.3897. Mean Confidence: 0.5677, Accuracy: 95.71%. This average mismatch is therefore underconfidence: predictions are correct much more often than their assigned maximum probabilities suggest.

Within the tested range, increasing the calibration sample size reduces the magnitude of the sigmoid-logit deterioration. Sigmoid-logit log loss at 90 observations is .3858, .2991 at 180 observations and .2523 at 360 observations. Respectively the difference from baseline is .2612, .1745 and .1277. All twenty partitions exhibit worse sigmoid-logit performance at every calibration size. Sample standard deviation of the increase is .0127 while the range of the increase for the full pool is .0981 to .1493. While increasing the size of the calibration pool tends to mitigate declines in probability quality in the test range; it does so without restoring probability quality to the baseline levels.

There are several reasons why the results cannot be easily interpreted. First, since they were obtained from finite data, there is limited information about the true underlying behavior. Second, if calibration-data scarcity alone explained the result, an adequately large calibration sample would need to remove the deterioration while competing structural explanations were controlled. However, even though there are significant limitations placed on the largest calibration set tested (i.e., 360 observations); continued degradation could represent any combination of representation issues, smoothing effects, estimator bias and/or residual calibration uncertainty. While the results illustrated a trend toward reduction in degradation; they did not illustrate a plateau, nor did they demonstrate that an infinitely sized sigmoid fit would fail in a similar manner.

3.3 Score representation and temperature scaling

Utilizing sigmoid-calibrated log odds rather than standard sigmoid logits produces better performance for all digit sizes. At 36 observations; utilizing log odds resulted in a log loss of .4840 versus .6152 with standard sigmoid logits. Log odds produced a loss of .1596 at 360 observations while standard sigmoid logits produced a loss of .2523. The full-pool paired deterioration for log odds was .0350 with a sample standard deviation of .0087 and exhibited a positive value for each of the twenty partitions. Sigmoid logits had an ECE value of .1340; log odds had an ECE value of .0608; and the baseline had an ECE value of .0233. Clearly, controlled substitution indicates that representation of inputs affect outputs; although damage occurs to varying extents.

However, a completely different story emerges with regard to temperature scaling. Average log loss:.1233 (change in log loss from paired data:-.0013±.0062) Sixteen out of twenty partitions showed improved log loss; four showed increased log loss. Similar to previously noted analyses, ECE values decreased to .0179. Accuracy was unchanged for each partition. Overall, we view this performance as being nearly equivalent to

baseline performance; with some overall improvements viewed as minor and not indicative of superior performance. The evaluated temperature range and objective remain part of this conclusion.

Temperature scaling resulted in an average log loss that increased to .2983 at 36 observations; and to .2141 at 90 observations. Means exhibited substantial variability across their respective splits. Full-pair deterioration for the smallest-size was between -.0119 and +.9160. Optimizers can aggressively sharpen predicted probabilities when small calibration sets contain few informative errors. Thus, we must determine what types of causal mechanisms result in boundary temperatures and loss dispersions representing practical instability during the calibration process; particularly when only one parameter is utilized to adjust the process.

3.4 Comparisons across multiclass datasets

Table 3 Paired probability-score changes using the full calibration pools

Dataset	Method	Δ log loss	Δ Brier	Δ ECE	LL harms
Digits	Sigmoid logits	+0.1277 ± 0.0127	+0.0489	+0.1107	20/20
Digits	Sigmoid log odds	+0.0350 ± 0.0087	+0.0084	+0.0375	20/20
Digits	Temperature	-0.0013 ± 0.0062	-0.0008	-0.0054	4/20
Iris	Sigmoid logits	+0.2876 ± 0.0206	+0.1739	+0.1361	20/20
Iris	Sigmoid log odds	+0.0509 ± 0.0152	+0.0273	+0.0292	20/20
Iris	Temperature	+0.1095 ± 0.3309	-0.0130	-0.0648	6/20
Wine	Sigmoid logits	+0.0997 ± 0.0148	+0.0356	+0.0829	20/20
Wine	Sigmoid log odds	+0.0882 ± 0.0199	+0.0304	+0.0669	20/20
Wine	Temperature	+0.1553 ± 0.3411	+0.0022	-0.0281	8/20

LL = logarithmic loss; ECE = confidence expected calibration error. Δ denotes corrected minus original, so negative values improve error scores. Log-loss dispersion is the sample standard deviation of paired changes. LL harms counts strictly positive changes; ties are excluded. Full pools contain 360 digits, 30 iris, and 36 wine observations.

Table 3 displays full-pool paired changes. Mean change, dispersion and observed ranges and harms counts for all calibration sizes – including those not included in the largest size pool – are presented in tables C7-C9.

Iris and wine also reproduced the sigmoid harm, but had different baseline confidence ECE. In fact, the ECE values were higher for Iris (0.1156) and Wine (0.0549), versus Digits' (ECE = 0.0233). These datasets extend the study beyond the low-ECE digits baseline. Sigmoid-logit log loss increased by 0.2876 and 0.0997 on iris and wine, respectively, with full pools of 30 and 36 calibration observations. Each of the partitions also demonstrated an increase in log loss. When utilizing the log-odds version instead, the changes for the log-odds versions were decreased to 0.0509 and 0.0882; these changes, like those for sigmoid logits, remained positive.

The effect of changing the score representation was larger on iris than on wine. Furthermore, calibration with the sigmoid function reduced accuracy (from 95.22% to 89.44%) compared to calibration with log-odds functions (which yielded 95.00%) in the iris dataset. Wine accuracies were close: 97.69% for the baseline model, 97.78%, and 97.50% for sigmoid logits and sigmoid log odds for wine. The different label implications explain why maintaining accuracy is important beyond simply ensuring quality of the probability output and show that sigmoid harm cannot always be interpreted as decreased confidence without changes to the predicted labels.

The tasks utilize differing metrics when applying temperature scaling. For example, at the full pool size, the ECE improved from 0.1156 to 0.0508 and the Brier score improved from 0.0872 to 0.0742, while the average log loss worsened from 0.1824 to 0.2919. On wine, ECE improved from 0.0549 to 0.0268 while log

loss worsened from 0.0834 to 0.2387. Wine Brier score increased from 0.0387 to 0.0409. Lower ECE therefore does not guarantee lower average log loss on the observed classes.

3.5 Partition variability and bin sensitivity

Mean temperature-based full-pool log-loss values are extremely variable with means of +0.1095 and standard deviations of 0.3309 for the iris and means of +0.1553 with standard deviations of 0.3411 for wine. Only six iris partitions and eight wine partitions became worse despite the unfavorable mean; however, the few large deteriorations outweighed the smaller improvements elsewhere. This trend is opposite to the consistently worsening log loss seen with sigmoid and demonstrates that both the magnitude and distribution of paired changes are relevant. The counts are descriptive and cannot be treated as independent binomial trials because the partitions overlap.

The direction of mean changes associated with full pools did not vary with the bin count change of ECE. Sigmoid-logit ECE increases were measured to be 0.1131, 0.1107, and 0.1076 for 5, 10, and 20 bins respectively on digits; temperature-scaling ECE changes were −0.0062, −0.0054, and −0.0048, respectively on digits as well. Iris and wine also retained the signs of their method-mean ECE changes across these bin counts. Full-pool sensitivities are listed in Table C10. Examples of additional directional patterns under various estimators of equal-width that were subjected to this sensitivity analysis are provided below; this analysis did not contribute to enhancing robustness with respect to class-wise evaluation, equally frequent bins or subgroup conditioning, and unseen observation.

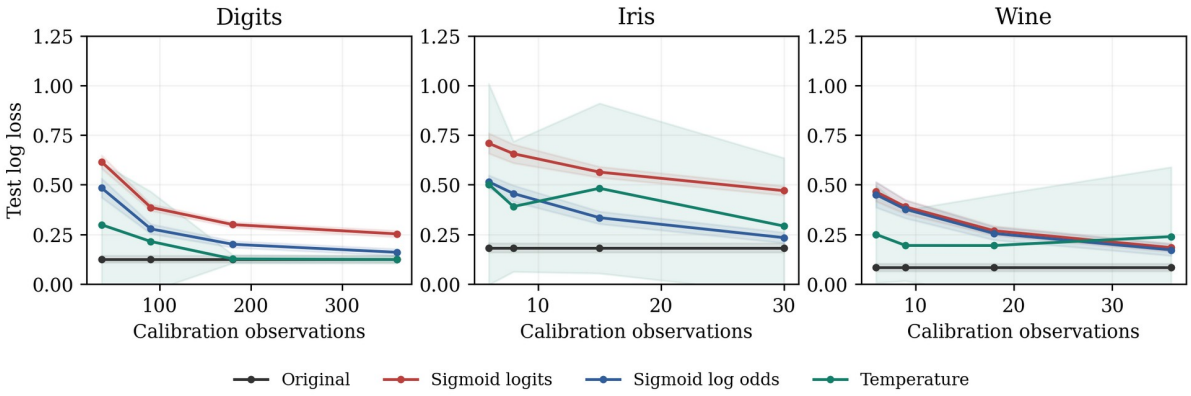


Figure 2 Calibration size and probability quality

Lines show means across twenty partitions; shading denotes ± one sample standard deviation and is not a confidence interval. The unchanged baseline repeats within each dataset. Vertical scales are comparable across the three panels.

4 Discussion

4.1 Interpretation of the controlled comparisons

An attempt was made to answer a broad question concerning the practicality of calibration through an empirical examination.

Smaller calibration budget resulted in larger detrimental impacts on class-wise sigmoid fitting. Reducing this impact could be achieved by altering the representations of scores relative to baseline models' fits. However, temperature scaling may stay near baseline probability performance if there is sufficient calibration data available for the digit task; however, temperature scaling can be unreliable with lower budgets or smaller tasks. Therefore, these results cannot be viewed as a comprehensive guide for determining which calibration procedures are most effective. Instead, these results provide guidelines for conducting implementation testing.

Log-odds substitution provides useful information since it maintains class-wise fitting mechanisms. Furthermore, log-odds substitution provides better evidence as to representation compared to evaluating unrelated algorithms.

4.2 Implications for evaluating probability corrections

Another limitation to this study involves a lack of a calibration selection process. A calibration-selection workflow was proposed to test both the candidate and unaltered predictors using the development data, and then to evaluate the selected method using an independent final test data set. Such a selection process was neither used nor tested in this experiment. As such, it provides tables that are directly comparable in order to indicate what was lacking in the "best" result from a test run - namely, a generalizable recommendation for deployment; selecting the best test result would reuse evaluation information. When only limited labeled data existed, an additional nested analysis would have been necessary in order to evaluate the reliability of selecting a beneficial calibration.

This study expands upon prior research into recalibration demonstrating damage due to variation in the amount of data available and representations of inputs. The study did not assert primacy for explaining the phenomenon under examination. Nor did it establish a new calibration algorithm. Rather than establishing a new calibration algorithm, the empirical contributions of this study include documenting that sigmoid damage still occurs at evaluated levels of data availability and that it is reduced through representation substitution, with temperature scaling providing a contrasting method. Replication of the examined phenomena across larger, independently collected datasets will be required in order to determine whether they support a more generally applicable explanation.

5 Limitations

Three of four small benchmark data sets used within this study were used in the targeted investigation. Iris and wine are particularly limiting to estimate calibration mappings as well as assess final-test probability quality. All characteristics vary simultaneously: number of classes; feature creation; data size; and task difficulty. Moreover, there is no analogous synthetic experiment that varies individual aspects of the above features. Hyperparameter searches were not conducted to test whether regularization changes the failure, and no external dataset was used to test transfer of the same fitted correction. Hence, observed behavior depends on the fitted default logistic regression model; how the features are standardized; splitting of the samples; and description of the implemented calibration.

Repeated runs are made in each partition, however, varying the seed generates no new subjects or authors. Descriptive standard deviations compute variability among partitions (non-inferential uncertainty regarding a target population). Stratification ignores possible groupings of writers and collections. Nested calibration prefix structures generate size-dependent relationships. These choices support controlled comparisons within partitions but limit population-level interpretation of the counts, ranges, and results.

There are many open questions in regards to diagnostic issues about why a particular problem occurred. For instance, how would a large multi-class data-set compare with regard to the calibrations budget beyond the maximum values. This would also be able to be evaluated using separate ablation studies on smoothing and normalizing. An additional potential score-shifting intervention could explore the softmax-invariant characteristics of classification outputs and utilize synthetic data with known conditional probabilities to discern between population miscalibration and sampling noise. Comparative evaluations would be possible to extend to additional correction mechanisms including vector-based or Dirichlet-based calibrations. Any additional evaluations would need to be substantiated based upon current evidence and future unknown data; they cannot simply be tacked on to provide a table of methodologies.

6 Conclusion

On digits, sigmoid harm decreases but remains across the increasing calibration sizes tested. On iris and wine, sigmoid harm also persists at the full calibration pools. Sigmoid harm can be decreased through the provision of sigmoid-fitting class-vs-rest log-odds, therefore reinforcing the value of representation. Temperature scaling exhibits similarities to the effective-digit baseline when utilizing the entire pool, although temperature scaling demonstrates increased sensitivity to scarcity and extreme-probability error occurrence. Calibration is considered to be an intervention whose usefulness is contingent upon the quality of representation, the type of data, and the selected probability measure. Therefore, this research supports further investigation and comparative assessments of methods instead of automatically recalibrating all correct classifiers.

Data and code availability

In addition to the grouped results provided in Appendices B and C, and in addition to the machine readable results provided on the index in Appendix D, the actual results and their subsequent code, along with 960 rows of subsequent evaluation, the pairs of modifications made, the full prediction vectors, the split indexes, and the environment version numbers are contained within the accompanying reproducibility package. All of these datasets are distributed through scikit-learn. The diagnostic utilizes scikit-learn 1.8.0; information regarding the environment that was utilized for computation is provided in the requirements file. No new subjects or primary observations existed. Neither a public repository identifier nor an archival Digital Object Identifier (DOI) has been asserted.

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Supplementary appendices

The appendices retain the complete supporting evidence. Table S1 identifies where to find definitions, original figures, detailed outcomes, and reconstruction records.

Table S1 Guide to the supplementary appendices

Appendix	Contents	Figures and tables
A	Abbreviations and mathematical notation	Tables A1–A2
B	Original benchmark and reliability evidence	Figures B1–B2; Tables B1–B6b
C	Full follow-up results and sensitivity analysis	Tables C1–C10
D	Data files, implementation, and reconstruction	Tables D1–D3

Appendix letters link to the corresponding sections in the Word document and PDF.

Appendix A Abbreviations and notation

A1 Abbreviations

Abbreviations defined in Table A1 represent abbreviations used in the text, equation identifiers, table header identifiers, reference identifiers, and supplementary file identifiers. Statistical dispersion is represented by standard deviation; not standard error. Additionally, all abbreviations are completely defined in the appropriate table footnotes to allow readers to interpret tables independently.

Table A1 Abbreviations and file format names

Abbreviation	Full form or designation	Use in this manuscript
API	Application programming interface	Public software interface used for the sigmoid equivalence check
BS	Brier score	Class errors are summed before averaging observations
CSV	Comma-separated values	Machine-readable tabular evidence
CV	Cross-validation	Suffix in the CalibratedClassifierCV class name
DOI	Digital object identifier	Persistent identifier in references
DOCX	Microsoft Word Open XML document format	File extension; a format designation rather than a literal acronym
ECE	Expected calibration error	Confidence-based, equal-width binned estimator in this study
JSON	JavaScript Object Notation	Structured prediction and split records
L-BFGS	Limited-memory Broyden–Fletcher–Goldfarb–Shanno	Optimization algorithm underlying the logistic solver
LL	Logarithmic loss	Mean negative log likelihood for the observed class
NLL	Negative log likelihood	Calibration objective; averaged over calibration observations
NumPy	Numerical Python	Numerical computing library; software name
PDF	Portable Document Format	Fixed-layout manuscript file
PMLR	Proceedings of Machine Learning Research	Publication series cited in the references
SD	Standard deviation	Sample dispersion across repeated partitions; not standard error
SHA-256	Secure Hash Algorithm 256-bit	Checksum algorithm used to identify the retained dataset snapshots
UCI	University of California Irvine	Dataset repository host
XML	Extensible Markup Language	Document representation referred to in the DOCX format designation
ZIP	ZIP archive format	Archive format name, not an acronym; contains code and evidence

The software module SciPy is a proper software name. Mathematical symbols, percent signs, and table indices are defined separately in Table A2.

A2 Mathematical notation

Table A2 provides a listing of mathematical notation used throughout the document. Zero-based indices from 0 to K−1 are utilized in the implementation arrays during the implementation of equations. Classifiers create a mapping of dataset labels to their respective ordered classes during evaluation of probability vectors. The indexing schemes outlined above do not alter any scores.

Table A2 Mathematical notation

Symbol	Definition
i; k; j	Observation index; focal class index; class-summation index
n; K	Number of evaluated observations; number of classes
n_{cal}	Calibration sample count in tables; the subscript distinguishes it from evaluation n
y_i; $\hat{y}_i$	Observed class; class with the largest predicted probability
p_{ik}	Original or currently evaluated probability of class k for observation i
c_i; z_i	Maximum class probability; indicator that the selected class is correct
s_{ik}	Multinomial logistic regression class score before softmax
r_{ik}	Class-versus-rest log odds calculated from original probabilities
q_{ik}; $\tilde{p}_{ik}$	Classwise sigmoid output before normalization; normalized corrected probability
a_k; b_k	Classwise sigmoid slope and intercept under the sign convention of Equation 6
T; $p_{ik}^{(T)}$	Positive temperature; probability after temperature scaling
σ	Logistic sigmoid function $\sigma(u) = 1/[1 + \exp(-u)]$
B; b; I_β	Number of confidence bins; bin index; set of observations in bin b
$\lvert I_\beta \rvert$	Number of observations in bin b; empty bins are omitted from Equation 3
1{condition}	Indicator equal to one if the condition holds and zero otherwise
Δ	Corrected-minus-original paired change within the same partition
n_+; n_-	Positive and negative classwise calibration counts for smoothed targets
C	Inverse regularization strength in the fitted logistic regression model
±	Mean plus or minus one sample standard deviation when used in tables
%	Percent; accuracy and label-change rates are reported on a 0–100 scale

The symbol b denotes a bin index in Equation 3 and a classwise intercept when written b_k in Equation 6. The distinct subscripts identify the contexts.

Appendix B Preliminary benchmark evidence

B1 Original reliability diagrams

Figures B1 and B2 provide illustrations of bin-summarization-by-binned-seed-zero experiments previously illustrated. Confidence calibration, not the risk of a known event such as malignancy, is depicted. Gray indicates equality in confidence and correctness. Illustrations above the line denote under-confidence while illustrations below the line denote over-confidence. Lines illustrating connections between occupied bins are solely illustrative and do not depict behavior between bins. Complete ten-seed bin file and Tables B6a and B6b and complete set of nine other seed files are included in Appendix D.

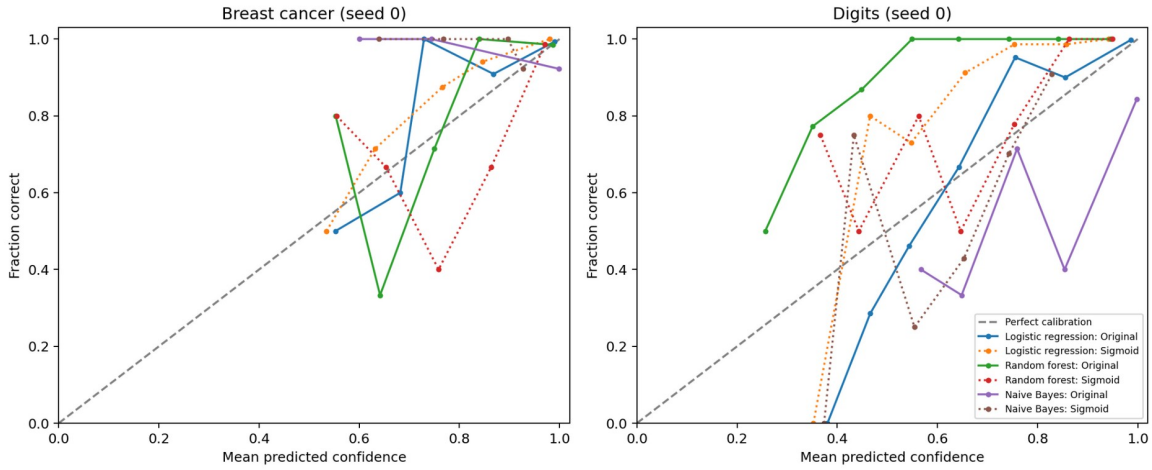


Figure B1 Unmodified original reliability plot

The original two-panel graph from the preliminary project is reproduced without changing its data or curves. Solid curves are original probabilities and dotted curves are sigmoid-calibrated probabilities, as identified in the original legend. All classifiers share each panel. Every point comes from seed zero; the underlying values are listed in Tables B6a and B6b.

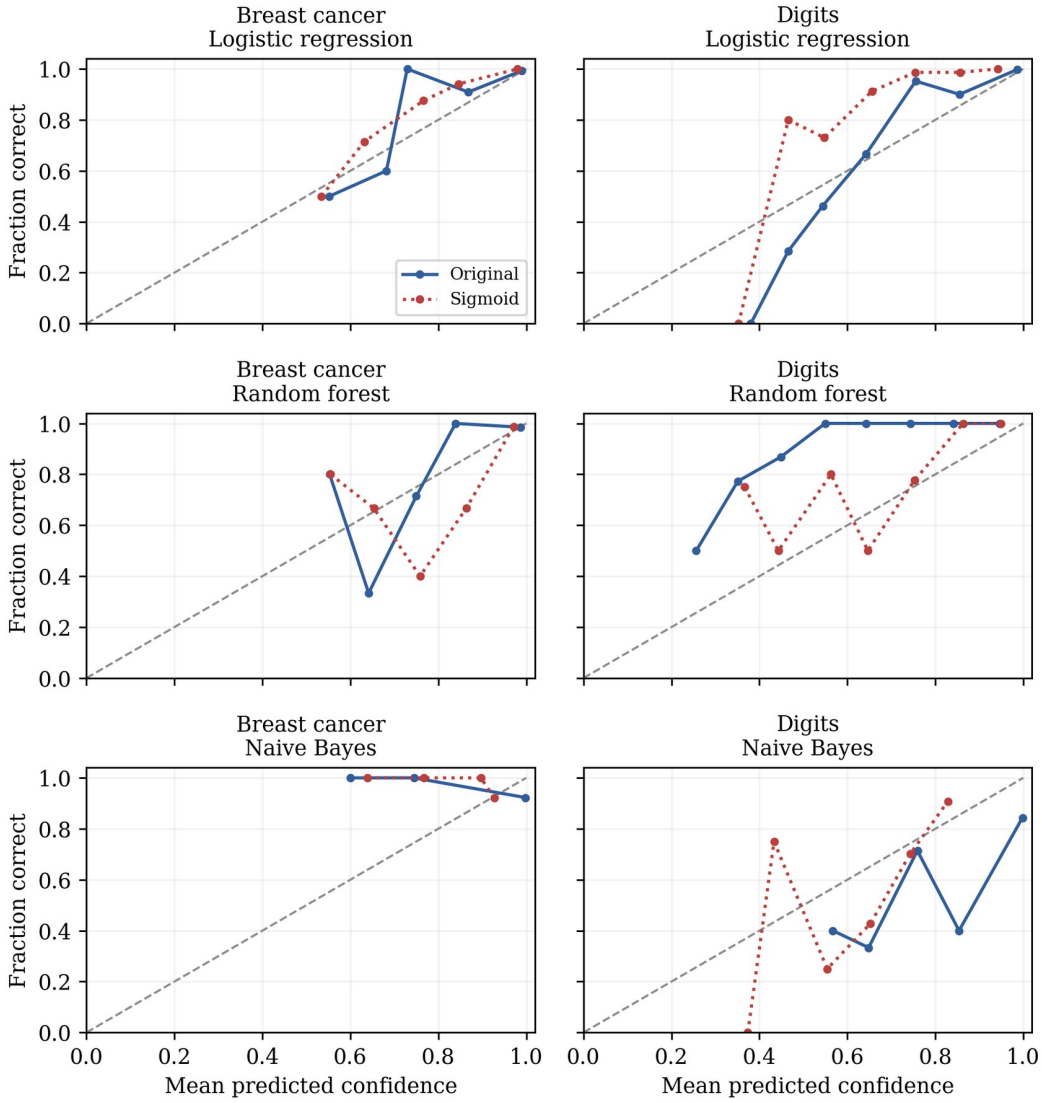


Figure B2 Original reliability results separated by classifier

The same seed-zero bin values as Figure B1, separated into six panels for readability. Original probabilities are blue solid lines and sigmoid-calibrated probabilities red dotted lines. The grey diagonal is agreement. Connecting lines do not estimate behavior between occupied bins. Tables B6a and B6b report each point and its sample count.

Complete two-panel illustration from our initial project is depicted once again without modifying its data or curves. Solid lines depict original probability while dotted lines depict sigmoid-calibrated probability as described in original legend. Each classifier uses each panel. Each point depicts an observation from seed zero; values tabulated in Tables B6a and B6b.

B2 Complete summaries and paired changes

The four major metrics are shown in tables B1 and B2, with all original means and sample standard deviations reported. The changes in both score and number of partitions are reported in Tables B3 and B4, respectively. Table B5 is a comparison of the mean confidence and accuracy. Each table has the same 10 partitions as the primary benchmark table. They are NOT independent population counts, but rather descriptive summaries of overlapping data.

Table B1 Complete Breast cancer benchmark summary

Classifier	Variant	Accuracy %	LL	BS	ECE
Logistic regression	Original	97.08 ± 0.95	0.0795 ± 0.0298	0.0439 ± 0.0157	0.0232 ± 0.0054
Logistic regression	Sigmoid	97.02 ± 1.15	0.0992 ± 0.0172	0.0501 ± 0.0130	0.0431 ± 0.0101
Random forest	Original	95.67 ± 1.70	0.1798 ± 0.1062	0.0690 ± 0.0163	0.0373 ± 0.0067
Random forest	Sigmoid	95.73 ± 1.85	0.1303 ± 0.0310	0.0684 ± 0.0204	0.0399 ± 0.0090
Naive Bayes	Original	93.86 ± 1.39	0.6656 ± 0.3032	0.1142 ± 0.0311	0.0594 ± 0.0156
Naive Bayes	Sigmoid	94.04 ± 1.37	0.2197 ± 0.0424	0.1096 ± 0.0267	0.0268 ± 0.0166

LL = logarithmic loss; BS = summed Brier score; ECE = ten-bin confidence expected calibration error. Values are mean ± sample standard deviation across ten partitions; accuracy dispersion is in percentage points.

Table B2 Complete Digits benchmark summary

Classifier	Variant	Accuracy %	LL	BS	ECE
Logistic regression	Original	96.43 ± 0.64	0.1264 ± 0.0154	0.0575 ± 0.0081	0.0240 ± 0.0060
Logistic regression	Sigmoid	95.83 ± 0.85	0.2559 ± 0.0169	0.1089 ± 0.0079	0.1332 ± 0.0122
Random forest	Original	96.93 ± 0.59	0.3621 ± 0.0116	0.1378 ± 0.0055	0.2339 ± 0.0100
Random forest	Sigmoid	96.94 ± 0.53	0.1381 ± 0.0119	0.0476 ± 0.0059	0.0621 ± 0.0063
Naive Bayes	Original	84.26 ± 2.04	2.9186 ± 0.4858	0.2984 ± 0.0394	0.1461 ± 0.0188
Naive Bayes	Sigmoid	84.61 ± 1.95	0.6857 ± 0.0652	0.2678 ± 0.0283	0.0725 ± 0.0357

LL = logarithmic loss; BS = summed Brier score; ECE = ten-bin confidence expected calibration error. Values are mean ± sample standard deviation across ten partitions; accuracy dispersion is in percentage points.

Table B3 Paired benchmark changes and improvement counts

Dataset	Classifier	Δ LL	Gains	Δ BS	Gains
Breast cancer	Logistic regression	+0.0196 ± 0.0153	1/10	+0.0061 ± 0.0037	1/10
Breast cancer	Random forest	-0.0494 ± 0.0918	5/10	-0.0006 ± 0.0053	4/10
Breast cancer	Naive Bayes	-0.4459 ± 0.2647	10/10	-0.0046 ± 0.0046	9/10
Digits	Logistic regression	+0.1295 ± 0.0119	0/10	+0.0515 ± 0.0064	0/10
Digits	Random forest	-0.2240 ± 0.0117	10/10	-0.0902 ± 0.0065	10/10
Digits	Naive Bayes	-2.2330 ± 0.4351	10/10	-0.0306 ± 0.0119	10/10

LL = logarithmic loss; BS = summed Brier score; ECE = confidence expected calibration error. Δ = calibrated minus original. Accuracy changes are percentage points. Gains counts strictly negative error-score changes or strictly positive accuracy changes; ties are excluded. Values are mean ± sample standard deviation of paired changes.

Table B4 Paired benchmark changes and improvement counts

Dataset	Classifier	Δ ECE	Gains	Δ accuracy	Gains
Breast cancer	Logistic regression	+0.0199 ± 0.0120	0/10	-0.0585 ± 0.6436	2/10
Breast cancer	Random forest	+0.0025 ± 0.0088	4/10	+0.0585 ± 0.4315	3/10
Breast cancer	Naive Bayes	-0.0326 ± 0.0250	9/10	+0.1754 ± 0.3947	2/10
Digits	Logistic regression	+0.1091 ± 0.0100	0/10	-0.5926 ± 0.3785	1/10
Digits	Random forest	-0.1718 ± 0.0097	10/10	+0.0185 ± 0.2217	2/10
Digits	Naive Bayes	-0.0736 ± 0.0295	10/10	+0.3519 ± 0.2822	9/10

LL = logarithmic loss; BS = summed Brier score; ECE = confidence expected calibration error. Δ = calibrated minus original. Accuracy changes are percentage points. Gains counts strictly negative error-score changes or strictly positive accuracy changes; ties are excluded. Values are mean ± sample standard deviation of paired changes.

Table B5 Confidence and accuracy in the preliminary benchmark

Dataset	Classifier	Original confidence %	Original accuracy %	Sigmoid confidence %	Sigmoid accuracy %
Breast cancer	Logistic regression	96.63	97.08	93.49	97.02
Breast cancer	Random forest	93.79	95.67	94.53	95.73
Breast cancer	Naive Bayes	99.39	93.86	92.64	94.04
Digits	Logistic regression	94.47	96.43	82.68	95.83
Digits	Random forest	73.62	96.93	91.37	96.94
Digits	Naive Bayes	98.70	84.26	81.42	84.61

Values are means across ten partitions. Matching mean confidence and accuracy does not establish calibration within bins or classes.

B3 Numerical values underlying the original diagrams

Each of the 67 bins illustrated in Figures 1, B1, and B2 are depicted in Tables B6a and B6b. These points do not represent confidence intervals. Bin-to-bin local variations may exist due to sparsity; therefore, these should be evaluated collectively with counts. All bounds include the lower end and exclude the upper end, except for interval 0.9–1.0 which contains both bounds. Each model variant group receives either 171 or 540 total number of observations for breast cancer or digit respectively.

Table B6a Breast cancer occupied confidence bins for seed zero

Classifier	Variant	Interval	Count	Confidence	Correct fraction
Logistic regression	Original	0.5–0.6	6	0.5524	0.5000
Logistic regression	Original	0.6–0.7	5	0.6821	0.6000
Logistic regression	Original	0.7–0.8	2	0.7293	1.0000
Logistic regression	Original	0.8–0.9	11	0.8682	0.9091
Logistic regression	Original	0.9–1.0	147	0.9901	0.9932
Logistic regression	Sigmoid	0.5–0.6	6	0.5346	0.5000
Logistic regression	Sigmoid	0.6–0.7	7	0.6316	0.7143
Logistic regression	Sigmoid	0.7–0.8	8	0.7661	0.8750
Logistic regression	Sigmoid	0.8–0.9	17	0.8465	0.9412
Logistic regression	Sigmoid	0.9–1.0	133	0.9805	1.0000
Random forest	Original	0.5–0.6	10	0.5530	0.8000
Random forest	Original	0.6–0.7	6	0.6417	0.3333
Random forest	Original	0.7–0.8	7	0.7500	0.7143
Random forest	Original	0.8–0.9	15	0.8393	1.0000
Random forest	Original	0.9–1.0	133	0.9872	0.9850
Random forest	Sigmoid	0.5–0.6	5	0.5547	0.8000
Random forest	Sigmoid	0.6–0.7	6	0.6535	0.6667
Random forest	Sigmoid	0.7–0.8	5	0.7585	0.4000
Random forest	Sigmoid	0.8–0.9	6	0.8636	0.6667
Random forest	Sigmoid	0.9–1.0	149	0.9711	0.9866
Naive Bayes	Original	0.6–0.7	1	0.6001	1.0000
Naive Bayes	Original	0.7–0.8	2	0.7447	1.0000
Naive Bayes	Original	0.9–1.0	168	0.9985	0.9226
Naive Bayes	Sigmoid	0.6–0.7	1	0.6392	1.0000
Naive Bayes	Sigmoid	0.7–0.8	2	0.7678	1.0000
Naive Bayes	Sigmoid	0.8–0.9	1	0.8969	1.0000
Naive Bayes	Sigmoid	0.9–1.0	167	0.9271	0.9222

Confidence and correct fraction are bin means on a 0–1 scale. Empty intervals are absent. Counts total 171 or 540 per classifier–variant group; all first-split plotted points are listed.

Table B6b Digits occupied confidence bins for seed zero

Classifier	Variant	Interval	Count	Confidence	Correct fraction
Logistic regression	Original	0.3–0.4	1	0.3805	0.0000
Logistic regression	Original	0.4–0.5	7	0.4660	0.2857
Logistic regression	Original	0.5–0.6	13	0.5437	0.4615
Logistic regression	Original	0.6–0.7	12	0.6429	0.6667
Logistic regression	Original	0.7–0.8	21	0.7553	0.9524
Logistic regression	Original	0.8–0.9	30	0.8551	0.9000
Logistic regression	Original	0.9–1.0	456	0.9863	0.9978
Logistic regression	Sigmoid	0.3–0.4	4	0.3525	0.0000
Logistic regression	Sigmoid	0.4–0.5	10	0.4657	0.8000
Logistic regression	Sigmoid	0.5–0.6	37	0.5476	0.7297
Logistic regression	Sigmoid	0.6–0.7	57	0.6556	0.9123
Logistic regression	Sigmoid	0.7–0.8	74	0.7540	0.9865
Logistic regression	Sigmoid	0.8–0.9	148	0.8570	0.9865
Logistic regression	Sigmoid	0.9–1.0	210	0.9420	1.0000
Random forest	Original	0.2–0.3	8	0.2562	0.5000
Random forest	Original	0.3–0.4	22	0.3509	0.7727
Random forest	Original	0.4–0.5	38	0.4487	0.8684

Classifier	Variant	Interval	Count	Confidence	Correct fraction
Random forest	Original	0.5–0.6	56	0.5495	1.0000
Random forest	Original	0.6–0.7	63	0.6421	1.0000
Random forest	Original	0.7–0.8	102	0.7425	1.0000
Random forest	Original	0.8–0.9	113	0.8410	1.0000
Random forest	Original	0.9–1.0	138	0.9463	1.0000
Random forest	Sigmoid	0.3–0.4	4	0.3665	0.7500
Random forest	Sigmoid	0.4–0.5	6	0.4431	0.5000
Random forest	Sigmoid	0.5–0.6	5	0.5627	0.8000
Random forest	Sigmoid	0.6–0.7	10	0.6466	0.5000
Random forest	Sigmoid	0.7–0.8	18	0.7532	0.7778
Random forest	Sigmoid	0.8–0.9	46	0.8630	1.0000
Random forest	Sigmoid	0.9–1.0	451	0.9496	1.0000
Naive Bayes	Original	0.5–0.6	5	0.5671	0.4000
Naive Bayes	Original	0.6–0.7	6	0.6484	0.3333
Naive Bayes	Original	0.7–0.8	7	0.7591	0.7143
Naive Bayes	Original	0.8–0.9	5	0.8540	0.4000
Naive Bayes	Original	0.9–1.0	517	0.9980	0.8433
Naive Bayes	Sigmoid	0.3–0.4	1	0.3738	0.0000
Naive Bayes	Sigmoid	0.4–0.5	4	0.4333	0.7500
Naive Bayes	Sigmoid	0.5–0.6	4	0.5543	0.2500
Naive Bayes	Sigmoid	0.6–0.7	7	0.6528	0.4286
Naive Bayes	Sigmoid	0.7–0.8	154	0.7428	0.7013
Naive Bayes	Sigmoid	0.8–0.9	370	0.8286	0.9081

Confidence and correct fraction are bin means on a 0–1 scale. Empty intervals are absent. Counts total 171 or 540 per classifier–variant group; all first-split plotted points are listed.

Appendix C Complete follow up summaries

C1 Outcomes across calibration sizes

In addition to the previously reported datasets, we also present the results of all studied calibration methods in terms of four primary metrics in Tables C1, C3 and C5 and the associated confidence levels and label updates in Tables C2, C4 and C6. The results are presented as mean ± SD for 20 different partitions. There is a calibration count defined for each of the conditions examined. The base-line rows appear repeatedly across the different conditions that were examined. These represent comparative base-lines and do not represent additional fitted models.

C1.1 Digits

Table C1 Digits complete metric summaries

n_{cal}	Variant	Accuracy %	LL	BS	ECE
36	Original	96.36 ± 0.65	0.1246 ± 0.0172	0.0564 ± 0.0072	0.0233 ± 0.0056
36	Sigmoid logits	95.71 ± 1.03	0.6152 ± 0.0335	0.2624 ± 0.0169	0.3897 ± 0.0226
36	Sigmoid log odds	96.09 ± 0.68	0.4840 ± 0.0479	0.1862 ± 0.0226	0.3004 ± 0.0323
36	Temperature	96.36 ± 0.65	0.2983 ± 0.3121	0.0617 ± 0.0102	0.0335 ± 0.0331
90	Original	96.36 ± 0.65	0.1246 ± 0.0172	0.0564 ± 0.0072	0.0233 ± 0.0056
90	Sigmoid logits	95.76 ± 0.89	0.3858 ± 0.0171	0.1550 ± 0.0082	0.2410 ± 0.0135
90	Sigmoid log odds	96.27 ± 0.55	0.2775 ± 0.0246	0.1020 ± 0.0101	0.1580 ± 0.0194
90	Temperature	96.36 ± 0.65	0.2141 ± 0.2513	0.0578 ± 0.0070	0.0248 ± 0.0182
180	Original	96.36 ± 0.65	0.1246 ± 0.0172	0.0564 ± 0.0072	0.0233 ± 0.0056
180	Sigmoid logits	96.01 ± 0.77	0.2991 ± 0.0136	0.1215 ± 0.0073	0.1745 ± 0.0117
180	Sigmoid log odds	96.38 ± 0.64	0.1999 ± 0.0165	0.0764 ± 0.0072	0.0962 ± 0.0114
180	Temperature	96.36 ± 0.65	0.1260 ± 0.0211	0.0560 ± 0.0074	0.0192 ± 0.0103
360	Original	96.36 ± 0.65	0.1246 ± 0.0172	0.0564 ± 0.0072	0.0233 ± 0.0056
360	Sigmoid logits	96.00 ± 0.62	0.2523 ± 0.0165	0.1053 ± 0.0086	0.1340 ± 0.0105
360	Sigmoid log odds	96.44 ± 0.65	0.1596 ± 0.0172	0.0648 ± 0.0076	0.0608 ± 0.0082
360	Temperature	96.36 ± 0.65	0.1233 ± 0.0193	0.0556 ± 0.0073	0.0179 ± 0.0086

n calibration = calibration observations; LL = logarithmic loss; BS = summed Brier score; ECE = ten-bin confidence expected calibration error. Values are mean ± sample standard deviation over twenty partitions.

Table C2 Digits confidence and label changes

n_{cal}	Variant	Mean confidence %	Changed labels %
36	Original	94.66 ± 0.41	0.00 ± 0.00
36	Sigmoid logits	56.77 ± 1.88	2.24 ± 1.08
36	Sigmoid log odds	66.12 ± 3.08	1.69 ± 0.90
36	Temperature	95.58 ± 4.43	0.00 ± 0.00
90	Original	94.66 ± 0.41	0.00 ± 0.00
90	Sigmoid logits	71.79 ± 1.20	1.97 ± 0.77
90	Sigmoid log odds	80.57 ± 1.86	1.15 ± 0.55
90	Temperature	95.81 ± 2.64	0.00 ± 0.00
180	Original	94.66 ± 0.41	0.00 ± 0.00
180	Sigmoid logits	78.71 ± 0.90	1.71 ± 0.53
180	Sigmoid log odds	86.89 ± 1.18	0.74 ± 0.41
180	Temperature	95.79 ± 1.51	0.00 ± 0.00
360	Original	94.66 ± 0.41	0.00 ± 0.00
360	Sigmoid logits	82.80 ± 1.00	1.52 ± 0.44
360	Sigmoid log odds	90.59 ± 0.94	0.70 ± 0.35
360	Temperature	95.71 ± 1.19	0.00 ± 0.00

n calibration = calibration observations. Values are mean ± sample standard deviation over twenty partitions. Changed labels denotes disagreement with the same fitted original classifier, not disagreement with the observed class.

C1.2 Iris

Table C3 Iris complete metric summaries

n_{cal}	Variant	Accuracy %	LL	BS	ECE
6	Original	95.22 ± 2.42	0.1824 ± 0.0240	0.0872 ± 0.0204	0.1156 ± 0.0213
6	Sigmoid logits	75.00 ± 6.84	0.7086 ± 0.0511	0.4028 ± 0.0278	0.2717 ± 0.0724
6	Sigmoid log odds	94.11 ± 3.25	0.5147 ± 0.0328	0.2607 ± 0.0209	0.3270 ± 0.0389
6	Temperature	95.22 ± 2.42	0.5022 ± 0.5062	0.0886 ± 0.0439	0.0473 ± 0.0222
8	Original	95.22 ± 2.42	0.1824 ± 0.0240	0.0872 ± 0.0204	0.1156 ± 0.0213
8	Sigmoid logits	87.11 ± 10.19	0.6565 ± 0.0470	0.3779 ± 0.0320	0.3365 ± 0.0931
8	Sigmoid log odds	93.67 ± 3.62	0.4558 ± 0.0412	0.2273 ± 0.0270	0.2851 ± 0.0346
8	Temperature	95.22 ± 2.42	0.3900 ± 0.3282	0.0866 ± 0.0449	0.0489 ± 0.0248
15	Original	95.22 ± 2.42	0.1824 ± 0.0240	0.0872 ± 0.0204	0.1156 ± 0.0213
15	Sigmoid logits	85.78 ± 5.60	0.5638 ± 0.0272	0.3205 ± 0.0177	0.2718 ± 0.0569
15	Sigmoid log odds	95.33 ± 2.59	0.3340 ± 0.0313	0.1633 ± 0.0183	0.2131 ± 0.0268
15	Temperature	95.22 ± 2.42	0.4820 ± 0.4283	0.0859 ± 0.0456	0.0518 ± 0.0216
30	Original	95.22 ± 2.42	0.1824 ± 0.0240	0.0872 ± 0.0204	0.1156 ± 0.0213
30	Sigmoid logits	89.44 ± 5.24	0.4700 ± 0.0235	0.2610 ± 0.0190	0.2516 ± 0.0531
30	Sigmoid log odds	95.00 ± 2.78	0.2333 ± 0.0265	0.1144 ± 0.0180	0.1447 ± 0.0251
30	Temperature	95.22 ± 2.42	0.2919 ± 0.3433	0.0742 ± 0.0439	0.0508 ± 0.0218

n calibration = calibration observations; LL = logarithmic loss; BS = summed Brier score; ECE = ten-bin confidence expected calibration error. Values are mean ± sample standard deviation over twenty partitions.

Table C4 Iris confidence and label changes

n_{cal}	Variant	Mean confidence %	Changed labels %
6	Original	85.89 ± 1.11	0.00 ± 0.00
6	Sigmoid logits	54.56 ± 3.79	24.00 ± 6.98
6	Sigmoid log odds	61.58 ± 2.09	2.44 ± 3.52
6	Temperature	99.14 ± 2.15	0.00 ± 0.00
8	Original	85.89 ± 1.11	0.00 ± 0.00
8	Sigmoid logits	55.21 ± 3.86	11.89 ± 9.60
8	Sigmoid log odds	65.81 ± 2.46	6.00 ± 4.27
8	Temperature	98.56 ± 2.37	0.00 ± 0.00
15	Original	85.89 ± 1.11	0.00 ± 0.00
15	Sigmoid logits	60.86 ± 2.76	13.22 ± 5.37
15	Sigmoid log odds	74.34 ± 2.15	2.33 ± 1.97
15	Temperature	97.84 ± 2.60	0.00 ± 0.00
30	Original	85.89 ± 1.11	0.00 ± 0.00
30	Sigmoid logits	65.81 ± 2.12	10.00 ± 4.53
30	Sigmoid log odds	82.05 ± 1.66	1.11 ± 1.69
30	Temperature	96.95 ± 2.55	0.00 ± 0.00

n calibration = calibration observations. Values are mean ± sample standard deviation over twenty partitions. Changed labels denotes disagreement with the same fitted original classifier, not disagreement with the observed class.

C1.3 Wine

Table C5 Wine complete metric summaries

n_{cal}	Variant	Accuracy %	LL	BS	ECE
6	Original	97.69 ± 1.58	0.0834 ± 0.0198	0.0387 ± 0.0131	0.0549 ± 0.0121
6	Sigmoid logits	95.83 ± 2.87	0.4646 ± 0.0485	0.2254 ± 0.0328	0.3182 ± 0.0335
6	Sigmoid log odds	96.48 ± 1.98	0.4502 ± 0.0644	0.2131 ± 0.0417	0.3093 ± 0.0435
6	Temperature	97.69 ± 1.58	0.2500 ± 0.2443	0.0449 ± 0.0282	0.0257 ± 0.0160
9	Original	97.69 ± 1.58	0.0834 ± 0.0198	0.0387 ± 0.0131	0.0549 ± 0.0121
9	Sigmoid logits	96.48 ± 1.46	0.3888 ± 0.0356	0.1779 ± 0.0236	0.2760 ± 0.0231
9	Sigmoid log odds	96.48 ± 1.69	0.3763 ± 0.0451	0.1677 ± 0.0282	0.2661 ± 0.0250
9	Temperature	97.69 ± 1.58	0.1939 ± 0.1788	0.0417 ± 0.0236	0.0254 ± 0.0155
18	Original	97.69 ± 1.58	0.0834 ± 0.0198	0.0387 ± 0.0131	0.0549 ± 0.0121
18	Sigmoid logits	96.94 ± 1.73	0.2683 ± 0.0237	0.1128 ± 0.0155	0.1914 ± 0.0224
18	Sigmoid log odds	96.85 ± 1.60	0.2545 ± 0.0328	0.1045 ± 0.0186	0.1794 ± 0.0212
18	Temperature	97.69 ± 1.58	0.1936 ± 0.2532	0.0418 ± 0.0250	0.0264 ± 0.0157
36	Original	97.69 ± 1.58	0.0834 ± 0.0198	0.0387 ± 0.0131	0.0549 ± 0.0121
36	Sigmoid logits	97.78 ± 1.76	0.1830 ± 0.0213	0.0743 ± 0.0138	0.1378 ± 0.0185
36	Sigmoid log odds	97.50 ± 1.83	0.1716 ± 0.0291	0.0691 ± 0.0169	0.1218 ± 0.0206
36	Temperature	97.69 ± 1.58	0.2387 ± 0.3503	0.0409 ± 0.0260	0.0268 ± 0.0160

n calibration = calibration observations; LL = logarithmic loss; BS = summed Brier score; ECE = ten-bin confidence expected calibration error. Values are mean ± sample standard deviation over twenty partitions.

Table C6 Wine confidence and label changes

n_{cal}	Variant	Mean confidence %	Changed labels %
6	Original	94.06 ± 1.11	0.00 ± 0.00
6	Sigmoid logits	64.43 ± 2.91	3.43 ± 3.31
6	Sigmoid log odds	65.70 ± 4.21	2.41 ± 2.26
6	Temperature	99.31 ± 1.56	0.00 ± 0.00
9	Original	94.06 ± 1.11	0.00 ± 0.00
9	Sigmoid logits	69.29 ± 2.28	2.59 ± 1.52
9	Sigmoid log odds	70.53 ± 2.98	1.85 ± 1.80
9	Temperature	98.85 ± 1.91	0.00 ± 0.00
18	Original	94.06 ± 1.11	0.00 ± 0.00
18	Sigmoid logits	78.32 ± 1.61	2.31 ± 1.58
18	Sigmoid log odds	79.70 ± 2.24	1.67 ± 1.58
18	Temperature	98.78 ± 1.66	0.00 ± 0.00
36	Original	94.06 ± 1.11	0.00 ± 0.00
36	Sigmoid logits	85.00 ± 1.48	1.30 ± 1.22
36	Sigmoid log odds	86.23 ± 1.95	1.02 ± 1.27
36	Temperature	98.70 ± 1.58	0.00 ± 0.00

n calibration = calibration observations. Values are mean ± sample standard deviation over twenty partitions. Changed labels denotes disagreement with the same fitted original classifier, not disagreement with the observed class.

C2 Paired log loss changes

Tables C7-C9 show corrected versus original log loss changes for all calibrated methods and sample sizes. Any decline toward the "worse" compared to the original represents an adverse change. Tie observations should NOT be included in your overall tally of adverse changes. The minima and maxima observed were not provided to give confidence limits. They are intended only to illustrate the consistently worsening sigmoid log loss and highly dispersed temperature values.

Table C7 Digits paired log loss changes

n_{cal}	Variant	Δ LL	Minimum	Maximum	Harms
36	Sigmoid logits	+0.4906 ± 0.0363	+0.4344	+0.6019	20/20
36	Sigmoid log odds	+0.3594 ± 0.0533	+0.2847	+0.5137	20/20
36	Temperature	+0.1737 ± 0.3178	-0.0119	+0.9160	13/20
90	Sigmoid logits	+0.2612 ± 0.0182	+0.2319	+0.3108	20/20
90	Sigmoid log odds	+0.1529 ± 0.0256	+0.1178	+0.2286	20/20
90	Temperature	+0.0895 ± 0.2559	-0.0098	+0.9161	13/20
180	Sigmoid logits	+0.1745 ± 0.0140	+0.1475	+0.2055	20/20
180	Sigmoid log odds	+0.0753 ± 0.0132	+0.0592	+0.1112	20/20
180	Temperature	+0.0014 ± 0.0089	-0.0106	+0.0203	9/20
360	Sigmoid logits	+0.1277 ± 0.0127	+0.0981	+0.1493	20/20
360	Sigmoid log odds	+0.0350 ± 0.0087	+0.0250	+0.0657	20/20
360	Temperature	-0.0013 ± 0.0062	-0.0110	+0.0142	4/20

LL = logarithmic loss; Δ = corrected minus original. Negative changes improve the score. Values are mean ± sample standard deviation of paired changes. Minima and maxima are observed values, not confidence limits; harms count strictly positive changes.

Table C8 Iris paired log loss changes

n_{cal}	Variant	Δ LL	Minimum	Maximum	Harms
6	Sigmoid logits	+0.5262 ± 0.0461	+0.4535	+0.6553	20/20
6	Sigmoid log odds	+0.3324 ± 0.0352	+0.2727	+0.3928	20/20
6	Temperature	+0.3198 ± 0.4912	-0.1541	+1.9464	14/20
8	Sigmoid logits	+0.4741 ± 0.0495	+0.3959	+0.6197	20/20
8	Sigmoid log odds	+0.2734 ± 0.0372	+0.2121	+0.3586	20/20
8	Temperature	+0.2077 ± 0.3249	-0.1542	+1.1390	13/20
15	Sigmoid logits	+0.3815 ± 0.0279	+0.3179	+0.4294	20/20
15	Sigmoid log odds	+0.1516 ± 0.0229	+0.1134	+0.1836	20/20
15	Temperature	+0.2997 ± 0.4228	-0.1256	+1.1390	12/20
30	Sigmoid logits	+0.2876 ± 0.0206	+0.2326	+0.3172	20/20
30	Sigmoid log odds	+0.0509 ± 0.0152	+0.0299	+0.0786	20/20
30	Temperature	+0.1095 ± 0.3309	-0.1249	+0.7656	6/20

LL = logarithmic loss; Δ = corrected minus original. Negative changes improve the score. Values are mean ± sample standard deviation of paired changes. Minima and maxima are observed values, not confidence limits; harms count strictly positive changes.

Table C9 Wine paired log loss changes

n_{cal}	Variant	Δ LL	Minimum	Maximum	Harms
6	Sigmoid logits	+0.3812 ± 0.0482	+0.3309	+0.5234	20/20
6	Sigmoid log odds	+0.3668 ± 0.0652	+0.2667	+0.5546	20/20
6	Temperature	+0.1666 ± 0.2300	-0.0637	+0.7162	14/20
9	Sigmoid logits	+0.3054 ± 0.0322	+0.2654	+0.4072	20/20
9	Sigmoid log odds	+0.2930 ± 0.0436	+0.2317	+0.4343	20/20
9	Temperature	+0.1106 ± 0.1729	-0.0637	+0.4043	12/20
18	Sigmoid logits	+0.1849 ± 0.0239	+0.1533	+0.2543	20/20
18	Sigmoid log odds	+0.1712 ± 0.0332	+0.1265	+0.2593	20/20
18	Temperature	+0.1103 ± 0.2444	-0.0642	+0.6590	8/20
36	Sigmoid logits	+0.0997 ± 0.0148	+0.0775	+0.1284	20/20
36	Sigmoid log odds	+0.0882 ± 0.0199	+0.0480	+0.1269	20/20
36	Temperature	+0.1553 ± 0.3411	-0.0642	+1.2215	8/20

LL = logarithmic loss; Δ = corrected minus original. Negative changes improve the score. Values are mean ± sample standard deviation of paired changes. Minima and maxima are observed values, not confidence limits; harms count strictly positive changes.

C3 Sensitivity to confidence binning

Table C10 compares three possible numbers of confidence bins (5, 10, and 20) equally spaced bins used over the entire collection of calibrations. The reported comparisons were paired within the partitions and originated from the same corrected probability vectors. Negative changes indicate lower estimated confidence ECE. Stability in such scenarios does not imply that the calibration is valid across subgroups, populations or classes.

Table C10 Full pool sensitivity to confidence bin count

Dataset	Variant	Δ ECE 5 bins	Δ ECE 10 bins	Δ ECE 20 bins
Digits	Sigmoid logits	+0.1131 ± 0.0105	+0.1107 ± 0.0113	+0.1076 ± 0.0114
Digits	Sigmoid log odds	+0.0395 ± 0.0074	+0.0375 ± 0.0086	+0.0356 ± 0.0079
Digits	Temperature	-0.0062 ± 0.0096	-0.0054 ± 0.0088	-0.0048 ± 0.0078
Iris	Sigmoid logits	+0.1308 ± 0.0567	+0.1361 ± 0.0536	+0.1319 ± 0.0544
Iris	Sigmoid log odds	+0.0244 ± 0.0293	+0.0292 ± 0.0318	+0.0274 ± 0.0301
Iris	Temperature	-0.0677 ± 0.0286	-0.0648 ± 0.0285	-0.0734 ± 0.0260
Wine	Sigmoid logits	+0.0865 ± 0.0204	+0.0829 ± 0.0183	+0.0802 ± 0.0145
Wine	Sigmoid log odds	+0.0691 ± 0.0191	+0.0669 ± 0.0183	+0.0664 ± 0.0189
Wine	Temperature	-0.0252 ± 0.0190	-0.0281 ± 0.0172	-0.0315 ± 0.0132

ECE = confidence expected calibration error; Δ = corrected minus original. Values are mean ± sample standard deviation across twenty partitions. Bin counts change the estimator, not the predictions.

Appendix D Reproducibility records

D1 Supplementary files

There are supplementary evidence files accompanying this paper, listed in Table D1. The first phase of study contained ten seeds with 120 evaluation runs across the ten seeds, along with 674 complete bins. The second phase included 960 additional evaluation runs and 720 additional corrected-minus-original rows. All probability vectors were stored in compressed JSON format, whereas all tabular records are in CSV format. It is impractical to enumerate every single prediction vector due to its indexed nature; thus, there are files with indexes that contain the actual values, allowing separate re-calculation of the same.

Table D1 Evidence files in the accompanying package

File	Extent	Evidence supplied
original_benchmark/results.csv	120 rows	All preliminary model, partition, variant, and metric evaluations
original_benchmark/summary.csv	12 groups	Original mean and sample standard-deviation summaries
original_benchmark/reliability_bins.csv	674 rows	Occupied-bin counts and coordinates for all ten preliminary seeds
original_reliability_uploaded.png	2 panels	Unmodified retained original-run reliability graph reproduced in Figure B1
original_benchmark/reliability.png	2 panels	Original project copy of the same plotted reliability results
original_logistic_reliability.png	2 tasks	Figure 1 with first-split bin-count histograms
original_reliability_panels.png	6 panels	Figure B2 separating the original classifier comparisons
calibration_size.png	3 panels	Figure 2 comparing mean LL across calibration sizes
followup_results.csv	960 rows	Every follow-up metric evaluation, including matched baseline rows
followup_summary.csv	48 groups	Means and sample standard deviations for all follow-up conditions
paired_changes.csv	720 rows	Every corrected-minus-original follow-up change
predictions.json.gz	960 records	Test probabilities, labels, indices, and condition identifiers
splits.json	60 records	Fitting/test indices and complete nested calibration ordering
datasets/*.csv; datasets/dataset_metadata.json	4 datasets	Raw features, labels, ordered row indices, class counts, and file checksums
environment.json; requirements.txt	Version records	Recorded versions and pinned dependencies
investigate.py	Executable source	Follow-up models, correction methods, metrics, and consistency checks
original_benchmark/experiment.py	Executable source	Preliminary fitting and calibration procedure
original_benchmark/make_graph.py	Executable source	Original plot generation from exported bin values

LL = logarithmic loss. CSV = comma-separated values; JSON = JavaScript Object Notation. The .gz extension denotes gzip compression. Complete numerical records are retained in the files rather than rounded manuscript tables.

D2 Computational environment and implementation

All code for this project was written in Python version 3.12, and utilized packages that include scikit-learn version 1.8.0, numpy version 2.3.5, scipy version 1.17.0, and pandas version 2.2.3. The fitted logistic regressions use the lbfgs optimizer with default tolerance of 10^{-4}, a regularization factor of C = 1, and maximum of 3000 iterations. During the creation of the model only training data is standardized; after which the training-derived transformation is applied to calibration and test data. The temperature optimization

procedure optimizes the negative Log Likelihood (NLL) of the calibration set in log-temperature space, with a tolerance of 10^{-8}. This represents a scalar optimization problem with a bounding constraint on the solution variable. A version-pinned private sigmoid fitter was utilized, and its predictions were checked against the public calibration API.

A detailed description of the experimental design as well as allocation specifics for both phases are outlined in sections 2.1 and 2.3. Follow up seeds: 100-119 Calibration ordering: seed + 10,000 Fits/calibration/test indices may be located in Splits.json. Various assertions regarding mutually exclusive sets of data, normalized finiteness of probabilities being utilized, proximity of parameters to those created using the softmax function, as well as maintainability of their labeling capabilities under positive temperatures are made. While these assertions will serve to validate consistency amongst the calculations that were completed, they will offer no assurance regarding whether the statistical quality of this experiment is sufficient.

D3 Data dictionary and reconstruction

Table D2 contains mappings between columns that have been pulled from tables and corresponding variables discussed in the paper. The term "grouped summary" refers to a display summarizing arithmetic means and sample standard deviation's (with one degree of freedom removed) for the collected data. As specified above, matching dataset, seed and calibration fractions refer to changes prior to calculation of averages. Separate recalculation of each subsequent follow-up metric is feasible through application of the probability vectors. Recalculating the original bin file enables reproduction of the original graph and expected calibration error (ECE) values; however, none of the original probability vectors were saved by this script.

Table D2 Data dictionary for the exported evaluation records

Exported column or key	Definition
dataset; seed	Task name and random partition identifier
model; variant	Preliminary classifier family and original/sigmoid probability variant
method	Follow-up probability variant: original, sigmoid logits, sigmoid log odds, or temperature
fraction	Nominal fraction of the reserved calibration pool; subject to the two-per-class minimum
n_fit; n_calibration; n_test	Counts of model-fitting, calibration, and evaluation observations
accuracy	Mean correctness on the test partition, exported on a 0–1 scale
log_loss	Mean negative logarithm of the probability assigned to the observed label
brier	Squared class-probability errors summed over classes and averaged over observations
ece; ece5; ece10; ece20	Confidence expected calibration error with the stated equal-width bin count; original ece uses ten
mean_confidence	Average maximum predicted class probability
temperature	Fitted positive temperature for temperature scaling; absent for other variants
label_change_rate	Fraction of test labels that differ from the original fitted classifier prediction
delta_accuracy; delta_log_loss; delta_brier	Corrected-minus-original changes in the corresponding test measures
delta_ece5; delta_ece10; delta_ece20; delta_mean_confidence	Corrected-minus-original changes in confidence error and mean confidence
bin; count; confidence; accuracy	Original bin number, sample count, mean confidence, and correct fraction
train; test; calibration_order	Disjoint fitting/test index arrays and reserved calibration indices in nested sampling order
test_indices; labels; probabilities	Follow-up record indices, observed labels, and ordered class probability vectors

Metric exports use proportions rather than percentages. Tables convert accuracy and confidence to percent where marked. A grouped standard deviation uses the sample convention with denominator equal to the number of partitions minus one.

D4 Dataset snapshots and row indexing

Table D3 identifies all feature and label snapshots that were used during testing. Every row is assigned a row index; each file contains the feature columns in loader order and the label. The row_index column in each file corresponds to the row indices in splits.json and predictions.json.gz. Class labels, target names, class counts and SHA-256 hash sums for each of the snapshot files are contained in dataset_metadata.json. Native scale measurements remain unchanged; standardization remains recreated from each training model partition.

Table D3 Dataset snapshots supplied with the experiment

Dataset	Rows	Features	Classes	Fitting / pool / test	Snapshot file
Breast cancer	569	30	2	284 / 114 / 171	datasets/breast_cancer.csv
Digits	1797	64	10	897 / 360 / 540	datasets/digits.csv
Iris	150	4	3	75 / 30 / 45	datasets/iris.csv
Wine	178	13	3	88 / 36 / 54	datasets/wine.csv

Pool denotes the reserved calibration pool, before the nested subsets in the follow-up. Labels are the zero-based class codes used by scikit-learn. Metadata lists feature names, class counts, target names, and SHA-256 checksums. The original dataset records are public benchmark observations, not newly collected measurements.